GCE

Rewarding Learning

Chief Examiner's and Principal Moderator's Report

Digital Technology



Summer Series 2026

Foreword

This booklet outlines the performance of candidates in all aspects of this specification for the Summer 2026 series.

CCEA hopes that the Chief Examiner's and/or Principal Moderator's report(s) will be viewed as a helpful and constructive medium to further support teachers and the learning process.

This booklet forms part of the suite of support materials for the specification. Further materials are available from the specification's microsite on our website at www.ccea.org.uk.

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GCE DIGITAL TECHNOLOGY

Chief Examiner's Report

Subject Overview

As the assessment of this specification continues to move on from the recent challenges, it is important to acknowledge that the evidence from this series confirms the continuing excellent work undertaken by centres in preparing and supporting candidates for all four modules. The number of centres who have properly prepared their candidates for this specification continues to increase.

The performance of candidates on the individual modules is detailed below. Centres are urged to read previous Chief Examiner's Reports for GCE Digital Technology as there is still evidence that many candidates could benefit from a better appreciation of the challenges they face in trying to meet all the specification's assessment objectives, particularly AO3.

It is important to repeat some key points from previous reports.

Candidates must:

- Be familiar with Digital Technology terminology to properly understand the questions
- Must use appropriate Digital Technology terminology in their responses
- They must consider the context of each question, particularly when a scenario is presented

Assessment Unit AS 1 Approaches to Systems Development

Overview

This paper consisted of five, multipart questions. Most of the question parts were worth up to six marks. The exceptions were two question parts worth eight marks and ten marks respectively. The question paper was designed to assess all areas of the unit specification and incorporated two questions which assessed Quality of Written Communication, each of which were awarded a maximum of six marks.

The evidence indicated that candidates found the assessment content to be accessible in terms of content assessed and time allocated to the question paper meaning all candidates were able to engage meaningfully with the paper. While the standard of answers varied in accordance with candidate ability and knowledge of content, there is a clear indication that candidates still need to focus on integrating Digital Technology terminology into their responses to ensure higher marks are accessible, in particular for the questions designated as assessing Quality of Written Communication (QWC).

Q1 **(a)** **(i)** Most candidates could identify at least three relevant responsibilities of the systems analyst. However, a significant number of candidates included fact-finding which was given in the scenario. Another relatively common failing was the use of inaccurate Digital Technology terminology. For example, 'user needs' is not acceptable for 'user requirements' at this level.

(ii) & **(iii)** The overall level of response to these two related questions was high, but a common reason for candidates not gaining full marks was a failure to justify why the feature of interviews/questionnaires they identified was in fact an advantage/disadvantage. Some candidates described the advantages of using interviews instead of questionnaires by describing the disadvantages of questionnaires and vice versa.

(b) **(i)** The main differences between a context level data flow diagram and a Level 1 data flow diagram were accurately described by most candidates.

(ii) The overall quality of the Level 1 data flow diagram was excellent and many completely accurate and properly annotated diagrams were given. A small number of candidates was careless in their use of the proper annotation. However, the main reason for candidates losing marks was because they did not properly label the two subprocesses even though these were clearly identified in the question.

Q2 **(a)** **(i)** This question required candidates to be familiar with the network diagram node notation identified in the specification and the majority of them could for maximum marks. Most of the remainder of the candidates were not at all familiar with the notation. Of the candidates who were familiar with the notation, some were careless in the words they used. The notation is clear and succinct. Thus, 'latest end time' is not acceptable for 'latest finish time'.

(ii) The overall level of response to this question was very disappointing. The question was quite specific and required candidates to explain how the critical path assists in the management of a software project. Only a very small number could describe the critical path. Most responses consisted of very general statements about project management.

(b) The overall level of response to this question was very good with most candidates being able to describe a constraint and a risk. The standard of the examples provided by most candidates was also good. However, some examples were not appropriate at this level such as 'money', 'time' or 'someone leaving'.

(c) Most candidates were aware of the concept of version management in a Digital Technology context. However, the question referred specifically to source code but many candidates ignored this focus.

(d) **(i)** Only a very few candidates had any knowledge of process modelling as a stage in RAD.

(ii) This QWC question, purposely designed to identify the top candidates, required candidates to compare RAD and the waterfall method with respect to how each involves the end user, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. Most candidates were broadly familiar with both approaches to system development, although much more so with the waterfall approach rather than RAD. Candidates were better at describing the approaches generally rather than how each involved the end user which was the focus of the question. As expected, very few candidates met the strict criteria for a top-band response. The QWC assessment afforded by these banded response questions is equally as important as the assessment of the specification's content, a key point many candidates ignore to their disadvantage. It must be stressed again that candidates will not be rewarded in this type of question for lengthy, unstructured and poorly expressed responses.

Q3 **(a)** **(i)** Most candidates were properly familiar with the contents of a test plan and many complete responses were provided.

(ii) There was significant confusion among candidates about the difference between extreme and exceptional data. This problem then impacted on many of the examples provided resulting in a disappointing level of response overall. The correct use of appropriate Digital Technology terminology is very important. For example, 'upper/lower bound/ boundary' are acceptable. 'Highest, lowest, on the edge' are not.

(b) Most candidates were properly familiar with system testing and acceptance testing and some excellent descriptions were provided. However, the question focused on the purpose of each of these types of testing. Many candidates ignored this important point.

Q4 **(a)** The question concerned data conversion when an existing information system is replaced by a new information system. Many candidates ignored this context and described data conversion in general without addressing any of the specific factors which would be of relevance in this case. The result was that they could not access the full range of the marks.

(b) **(i)** Most candidates demonstrated a familiarity with a disaster recovery plan. However, the question required them to describe three considerations such a plan should address. Many candidates struggled to provide the detail indicated by the two-mark allocation for each consideration, particularly with regard to the Digital Technology terminology expected. A good illustration of this is the answer in the mark scheme for two marks – 'identify an alternative location at which the organisation can function until the disaster has been dealt with'. The response of many candidates could be typified as simply stating 'another place/location' which falls very far short of what is required at this level.

(ii) This QWC question, purposely designed to identify the top candidates, required candidates to describe the two methods of backup and explain how the data will subsequently be recovered, in a concise, focused answer using relevant Digital Technology terminology and with a high standard of QWC. Most candidates were broadly familiar with both methods of data backup, although many failed to distinguish clearly between them. In general, candidates struggled to describe the recovery methods which was the focus of the question. As expected, very few candidates met the strict criteria for a top-band response. The QWC assessment afforded by these banded response questions is equally as important as the assessment of the specification's content, a key point many candidates ignore to their disadvantage. It must be stressed again that candidates will not be rewarded in this type of question for lengthy, unstructured and poorly expressed responses.

Q5 **(a)** **(i)** The majority of candidates could identify at least three other IDE facilities. However, some candidates identified translation even though this was given in the preamble to the question.

(ii) The overall level of response to this question was very high and many candidates obtained maximum marks. Those candidates who did not obtain high marks often mixed up the two types of translation.

(b) This question split the candidates almost equally into three groups. One group demonstrated that they had little or no practical experience of programming. A second group had some programming experience but they did not really understand the fundamental IF THEN construct. The third group had practical experience of selection but not of a nested IF THEN ELSE. Only a small number of candidates gained significant marks for their algorithm.

(c) As in previous sessions, only a small percentage of candidates demonstrated the required familiarity with these fundamental OO concepts which enabled them to use appropriate OO terminology to obtain all or most of the available marks.

Assessment Unit AS 2 Fundamentals of Digital Technology

This paper contained five, multipart questions with marks available running from two marks to a maximum of six marks per question. The question paper was designed to assess all areas of the unit specification and incorporated three questions which assessed Quality of Written Communication, each of which were awarded a maximum of six marks.

Evidence indicated that candidates found the assessment content to be accessible in terms of content assessed and time allocated to the question paper meaning all candidates were able to engage meaningfully with the paper. While the standard of answers varied in accordance with candidate ability and knowledge of content there is a clear indication that candidates still need to focus on integrating Digital Technology terminology into responses to ensure higher mark bands, for example, in QWC questions remain accessible.

Q1 **(a)** Most candidates were able to correctly identify the smallest and largest decimal values that could be represented using four bits. A small number of candidates presented their response in binary as opposed to decimal format meaning marks could not be awarded.

(b) The majority of candidates were able to correctly identify the number of bits in 2 kilobytes of data with a variety of viable approaches being taken to completing the associated calculations.

(c) While most candidates were able to correctly identify the number of bits required to represent the decimal value presented in the question stem, some were not able to access all marks in this question part as they did not show working out as requested.

(d) It was pleasing to see working out well laid out by most candidates when answering this question which required conversion of a decimal value into an 8-bit binary number but not all candidates responded with an 8-bit binary number meaning not all marks were then available to them. Emphasis again should be placed on reading all instructions and question parts to ensure accessibility of all marks in each question in an examination paper.

(e) When converting the negative number presented in the question stem to 2's complement format some candidates used 2's complement binary headings and manual addition of values to -128 to convert the decimal number presented into 2's complement format. On some occasions, errors were made when adding 1 to the inverted representation. This meant these candidates could only be awarded part marks for the process and marks allocated to generating the correct response were missed.

Similarly for some candidates who took the approach where the positive binary alternative was produced, before inverting the bits and adding one, errors were made during the process of binary addition.

(f) Most candidates were able to successfully determine the magnitude of the number represented by the 2's complement binary value shown in the question stem. Some, however, could not be awarded full marks as they failed to include the negative sign in their response.

It is important that centres impress on candidates the need to show working out for such questions as this can prevent candidates being penalised for mistakes in calculations in earlier stages in formulating their response, meaning again only partial marks could be awarded. At all times, when calculations are required as part of a response to a question, candidates should be encouraged to take care when completing calculations in such questions in future examinations.

Q2 **(a)** **(i)** While candidates are able to successfully define the terms data and information, many still provide examples that do not relate to the context provided in the question stem. Candidates should be reminded of the need to reference question contexts when required as part of their overall response as not doing so can negatively impact on their total marks for a given question.

(ii) Candidates were in the main very successfully able to provide an appropriate example of a decision made using the context of the question stem. Where some candidates failed to achieve the maximum available marks was through lack of a detailed definition of the term knowledge.

(b) The focus of this question was on the difference between hash and control totals. Some candidates focused on the idea that the two terms related to control totals or the fact that they were both used as error detection mechanisms (both of which are similarities). Quite a large proportion of candidates were able to successfully identify the correct hash and control total from the examples provided but had difficulty providing an appropriate definition of each term. It was apparent however, that some candidates are still unable to distinguish between the two terms.

(c) Candidates who were awarded maximum marks in this question provided a clearly laid out response which showed all elements of working out in a systematic and well organised format. Some candidate responses showed that there was a clear understanding of the process involved in calculating a check digit but some confusion over the values of the weightings that needed to be applied to numbers during the execution of the process with some starting with a weighting of 1 being placed on the least significant bit. In questions such as this one where a clear process can be applied and where candidates are asked to show their working out, centres should impress upon candidates the importance of showing all steps clearly as marks are made available for the process being followed in addition to the final result.

(d) This question focused on the effectiveness of verification methods as a means of ensuring the correctness of data entered into a computer system. A prompt was also provided to encourage candidates to reference two different methods of verification. Some candidates discussed validation methods as opposed to verification methods. The majority of candidates were able to provide reasonable descriptions of two types of data verification but only a small proportion of candidates were able to comment on the effectiveness of verification methods as a means of ensuring correctness of data.

Q3 **(a)** **(i)** While most candidates were able to identify appropriate reasons for the selection of Computer B as opposed to Computer A, a large number of candidates found it was more difficult to provide a reason why this difference was important in the selection process. Some candidates also provide reasons which were based on assertion as opposed to the question stimulus, for example, referencing the storage provision. Centres should remind candidates of the need to consider only the information provided in the question stimulus when responding to questions of this nature.

(ii) The overall level of response to this question was high with most candidates being able to provide detailed explanations of the two terms provided to help explain the need for both RAM and secondary storage in a computer system.

(iii) Most candidates were able to provide at least two solid reasons why flash memory is suitable for use as a secondary storage medium. Some candidates provided responses such as light, small, fast but did not elaborate on these terms. Single word responses are not adequate when attempting to provide a reason in a question of this nature and candidates should be reminded of the need to provide further elaboration to help qualify their response.

(b) Most candidates were able to provide appropriately detailed descriptions of the terms provided showing there is an overall high level of understanding of the purpose of the components listed.

(c) A significant portion of candidates failed to correctly identify the components referenced in the descriptions listed in the table provided in this question part. Most were able to correctly identify the MDR and PC but struggled with the Control Bus description.

(d) This QWC question required candidates to show the difference between multiprocessing and multitasking operating systems. A large proportion of candidates struggled to clearly define the two terms clearly. Many were able to provide a detailed description of multitasking but had difficulty clearly outlining features of multiprocessing with appropriate detail. Most were able to provide reasonable examples of how multitasking could be used in computer systems but some found it more difficult to provide a suitable use or scenario where multiprocessing might be required. Candidates who were able to provide descriptions of both terms at times were only able to do so with limited technical terminology in their response and this limited their ability to access marks in the Band 2. Candidates should be reminded in QWC questions of the need to provide appropriate detail in their definitions or descriptions of the technologies identified in the question stem in addition to ensuring they provide an appropriate example to illustrate their response if they are to access marks in the higher mark band in such questions.

Q4 **(a)** Many candidates struggled with this question and provided general descriptions of different operating system types as opposed to focussing their response on the purpose of operating systems.

(b) Most candidates had a clear understanding of the main features of both utility applications listed in the question stem. There was a greater familiarity with archive software as opposed to file backup however.

(c) A large proportion of candidates struggled to clearly define the term forms dialogue interface with many only being awarded one out of the maximum two marks available. Candidates have a clear understanding of the term but struggled with finding the appropriate technical terminology to help correctly answer this question.

(d) Candidate responses to this question show there was a clear understanding of the difference between real-time and batch processing systems as most candidates were able to provide detailed technical responses related to describing both systems. Where some candidates struggled however was in explaining what made real-time processing more suitable to the context provided. Some candidates failed to link the processing system to the scenario provided while others focussed on booking systems which was not referenced in the scenario.

(e) **(i)** & **(ii)** Candidates in the main understand the difference between custom-built and off–the–shelf solutions and some candidates focussed on providing descriptions of the two solution types as opposed to detailing advantages of custom-built for Part i and advantages of off–the–shelf for Part (ii).

Some candidates failed to use the correct technical terminology in their references to the advantages of each solution type and this impacted on the overall marks awarded for their responses.

Q5 **(a)** It was pleasing to see a large number of candidates successfully identify which elements in the table correctly contained the domain name and country level domain in the examples shown from the CCEA fact file.

(b) The overall level of response to this question was disappointing. Only a small number of candidates were able to provide appropriate definitions of the terms domain name and URL. Responses tended to be lacking in technical terminology and showed some candidates were unable to distinguish clearly between the two terms. There were some instances where candidates were able to correctly provide correct examples to illustrate the difference but their associated descriptions of the terms lacked appropriate technical detail.

(c) Overall, the level of candidate response to this question was very high showing candidates have a clear understanding of how HTML tags can be used and modified to support changes to how multimedia application content is presented when displayed.

(d) Candidates in the main were able to draw upon their own experience of how the intranet is used in an organisation (school) to help provide an appropriate response to this question. Using a context which was familiar to candidates was designed to help maximise question accessibility. While many candidates were able to provide a suitable example of how an intranet could be used under each of the headings provided, many failed to provide appropriate elaboration meaning they were unable to access all marks available.

(e) In this QWC question, candidates were provided with a scenario and asked to use examples from the scenario to help distinguish between client-side processing and server-side processing. Many candidates failed to identify the key difference between both types of processing but a significant number of candidates were able to provide accurate examples which correctly showed there was an understanding of how/when each type of processing occurred in the scenario provided. Many, however, failed to use appropriate technical terminology to describe the two terms provided meaning their access to Band 2 marks was limited.

Assessment Unit A2 1 Information Systems

This paper consisted of six, multipart questions. Most of the question parts were worth from two to six marks. The exceptions were two question parts worth eight marks and nine marks respectively. The question paper was designed to assess all areas of the unit specification and incorporated three questions which assessed Quality of Written Communication, each of which were awarded a maximum of six marks.

The evidence indicated that candidates found the assessment content to be accessible in terms of content assessed and time allocated to the question paper meaning all candidates were able to engage meaningfully with the paper. While the standard of answers varied in accordance with candidate ability and knowledge of content there is a clear indication that candidates still need to focus on integrating Digital Technology terminology into their responses to ensure higher marks are accessible, in particular for the questions designated as assessing QWC.

Q1 **(a)** In general, candidates were familiar with the three network resources, particularly WAP and switched hub. The descriptions of IP address generally focused on the format of an IP address rather than its purpose in a network as was required.

(b) Only a very small number of candidates could identify even one type of server. When a correct type of server was identified, a proper description was very rare. Thus, a file server 'managed files'. Very few candidates obtained marks in this question.

(c) The majority of candidates were familiar with a ring network and some excellent descriptions of the role of the token in data transmission were provided. A number of candidates focused their answers on irrelevant features of a ring network such as adding new nodes or the effect of a cable break.

(d) This QWC question, purposely designed to identify the top candidates, required candidates to compare peer-to-peer networks and client-server networks with regard to how network resources are shared, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. Most candidates were broadly familiar with both networks, but a large number of candidates focused their responses entirely or almost entirely on their general features and ignored how resources are shared on each network. As expected, very few candidates met the strict criteria for a top-band response. The QWC assessment afforded by these banded response questions is equally as important as the assessment of the specification's content, a key point many candidates ignore to their disadvantage. It must be stressed again that candidates will not be rewarded in this type of question for lengthy, unstructured and poorly expressed responses.

Q2 **(a)** In general, candidates could not identify any elements defined by a communication protocol in spite of the guidance given in the question.

(b) **(i)** The majority of candidates could summarise the OSI model as required.

(ii) The overall level of response to this question was very high. Candidates were familiar with the two OSI layers and could describe them appropriately.

(iii) When asked to describe the OSI model, candidates frequently describe the OSI model correctly as a framework. This question required them to justify this statement but very few were able to do so.

(iv) Candidates were more familiar with the benefits of the OSI model and many complete and accurate responses were given.

(v) The overall level of response to this question was very good. Candidates were familiar with TCP/IP and Bluetooth protocols and most were awarded high marks.

Q3 **(a)** The overall level of response to this question was disappointing. While many candidates could attempt a brief explanation of an entity, most found it difficult to explain the term attribute. There was considerable confusion when candidates attempted to identify an example of either in the table.

(b) **(i) (ii) (iii)** The overall level of response to this question was good. Many candidates were adept at normalizing the data as required and many completely correct answers were given. However, as in previous series, a number of candidates ignored the requirement to use the appropriate notation, particularly the fundamental requirement that each entity must be identified by name and that the primary keys of each entity must be identified by the use of underlining or by the use of the # symbol. The notation is very clear and is not complicated. This weakness has been highlighted frequently in past Chief Examiner's Report.

(c) The overall level of response to this question was also good. Many complete and accurate answers were given.

(d) The high standard continued in this question. The overall standard of the entity-relationship models was very high. Many were completely correct. The high number of diagrams which complied completely with the required annotation continued this very positive trend in recent series.

Q4 **(a)** **(i)** The overall level of response to this question was much higher than when this area of the specification was assessed previously. Many candidates had a good understanding of the nature of both a logical data model and a physical data model.

(ii) The overall level of response to this question was very high. In general, candidates were familiar with the data dictionary and most could correctly identify at least four components.

(b) **(i) (ii) (iii)** The percentage of candidates who have practical experience of SQL continues to increase and many completely correct code segments were given. The question achieved the intended degree of discrimination. Creating and retrieving data proved straightforward to these candidates while updating a record was more of a challenge and provided the intended differentiation across the candidature.

(c) This QWC question, purposely designed to identify the top candidates, required candidates to evaluate the use of QBE as an alternative to SQL for creating database queries, in a concise, focused answer using relevant Digital Technology terminology and with a high standard of QWC. Most candidates were broadly familiar with both approaches, but a large number of candidates produced answers almost entirely composed of general features which failed to focus on database query production. As expected, very few candidates met the strict criteria for a top-band response. The QWC assessment afforded by these banded response questions is equally as important as the assessment of the specification's content, a key point many candidates ignore to their disadvantage. It must be stressed again that candidates will not be rewarded in this type of question for lengthy, unstructured and poorly expressed responses.

Q5 **(a)** Most candidates were familiar with the Turing Test but descriptions of how the test is carried out varied greatly in quality. Many candidates confused the three roles involved which suggested they had no real understanding or practical experience of what the test actually involves. The question concerned how the test determined if a machine is demonstrating human intelligence but many candidates did not address this properly.

(b) **(i)** The majority of candidates could describe the terms shell and fuzzy logic as applied to expert systems. However, many candidates described a general user interface, not that of an expert system.

(ii) The format of this question enabled those candidates who were properly familiar with expert systems to show their understanding with minimal writing effort. Many completely correct answers were given but the question also achieved the intended degree of discrimination.

(c) This QWC question, purposely designed to identify the top candidates, required candidates to use appropriate Digital Technology terminology to describe mobile phone communications, in a concise, focused answer using additional relevant DT terminology and with a high standard of QWC. The Digital Technology terms to be used were provided. Many candidates ignored these terms. Instead, they produced their own version of the sequence of events involved. As expected, very few candidates met the strict criteria for a top-band response. The QWC assessment afforded by these banded response questions is equally as important as the assessment of the specification's content, a key point many candidates ignore to their disadvantage. It must be stressed again that candidates will not be rewarded in this type of question for lengthy, unstructured and poorly expressed responses.

Q6 **(a)** **(i)** As in previous series when this part of the specification is assessed, many candidates struggled to explain the difference between data mining and big data.

(ii) This question referred to banks using data mining to help prevent fraud. Many candidates ignored one or both of these elements. Their general and unconnected references to data mining and fraud meant these candidates obtained few marks.

(b) **(i)** The overall level of response to this question was disappointing. Candidates were required to describe methods used by banks to ensure customer data is kept secure in the context of Data Protection legislation. Many candidates failed to reflect this focus and answers such as keeping doors locked and backing up data were common.

(ii) The overall level of response to this question was very high. The only significant weakness was a failure to use the language contained in the DP Act as is expected at this level.

(iii) & **(iv)** The overall level of response to this question was also very high.

(c) The overall level of response to this question was disappointing. Many candidates focused exclusively on provisions of the Computer Misuse Act itself, thus ignoring the fact that the question referred to steps a bank should take to ensure employee compliance.

Principal Moderator's Report

Assessment Unit A2 2 Application Development

Internal Assessment Overview

This is a piece of coursework submitted by Candidates based on a pre-released case study. In this series the case study was called BelTrails and it carries a 20% weighting of the overall A Level outcome. Candidates spend approximately 60 hours documenting and producing a relational database solution for BelTrails. Centres internally assess the candidates work using assessment indicators and submit outcomes to CCEA. Moderation of centre outcomes by CCEA then takes place.

Unit Overview

The pre-released case study for the 2026 series was entitled BelTrails. The overall standard of work in the samples submitted across centres was generally good. Most candidates produced complete database solutions which addressed the main requirements of the case study. Candidates were able to demonstrate a sound understanding of database concepts and were generally able to create relevant solutions containing tables, forms, queries, reports and macros. The majority of submissions were well presented and clearly organised in appropriate folders. Whilst candidates provided evidence of analysis, design, application development and documentation, the quality and depth of evidence in the solutions varied considerably.

Many candidates successfully implemented the core requirements specified in the case study. The candidates who achieved the higher mark bands in their submissions demonstrated a wider range of functionality, including more complex queries, calculated fields, improved navigation systems and more sophisticated reporting facilities. The candidates who achieved lower mark bands in their submissions included limited functionality only considering the basic requirements of the task, making limited use of the features available within Microsoft Access.

The application of the assessment criteria was generally consistent across the samples submitted.

In the majority of solutions the marks awarded reflected the quality of evidence presented. The overall standard of coursework was good and most candidates produced database applications that met the requirements of the BelTrails case study. Stronger candidates demonstrated clear links between analysis, design, application development and evaluation. Some candidates would benefit from providing more detailed explanations of their work and more evaluative commentary throughout their portfolio. The teacher comments included were helpful in clarifying and justifying why marks were awarded. Marking was within the agreed standard for most centres and internal standardisation procedures were evident across centres. In the majority of centres, candidates demonstrated a strong understanding of the requirements for this case study. The assessment criteria allowed learners of all ability levels to be assessed appropriately.

There still exists evidence within some samples of over-guidance by Centres. It would be beneficial to see these candidates taking a more individual approach rather than producing very similar material, which may suggest a template approach or over-guidance by the Centre.

Analysis

Candidates were able to choose a suitable scope to follow. There was a good mix of ordering and booking systems for BelTrails. In this section, candidates are required to produce detailed and relevant user requirements for the case study. It was pleasing to note that many candidates were able to conduct a high-level analysis of the case study and produce an appropriate set of detailed and justified user requirements. Some candidates included descriptions of user requirements but lacked sufficient detail when justifying why the requirements were needed for their solution. In some cases user requirements were identified but not subsequently linked to the overall system. Some DFD Level 1 diagrams lacked detail including names for data flows.

Design

The best submissions demonstrated effective use of relationships and referential integrity, appropriate validation and input controls, high quality forms and reports design, highly structured and complex query design and user-friendly navigation.

Candidates who attained higher marks in this section were able to produce detailed planning documentation based on their user requirements, including database normalisation, entity relationship diagrams, data dictionaries and screen designs. There was clear evidence in these cases that the planning documentation informed the development of their solution.

Database normalisation continues to challenge some candidates. Candidates are required to follow the correct processes for normalisation through the three stages. A stronger understanding of this key area of the specification would not only support the development of their solution but also improve performance in normalisation questions in the A2 examination. Candidates should create test plans linked to user requirements which will also help them in developing their application solution.

Application Development

The majority of candidates developed functioning database solutions linked to the requirements of the case study. The overall standard was very good with a range of ACCESS functionality used in solutions. Solutions were imaginative and creative including a variety of approaches ranging from an Ordering system, an Invoice system and a stock system. It was pleasing to see Candidates taking different approaches. A number of candidates relied heavily on screenshots to demonstrate their implementations. While screenshots can provide useful evidence, candidates should ensure that sufficient explanation accompanies them to demonstrate understanding during development.

The strongest testing sections contained evidence of normal, exceptional and extreme test data together with expected outcomes, actual outcomes and details of any corrective action taken.

In weaker portfolios, testing often consisted of a large number of screenshots with limited explanation. Some candidates demonstrated that database functionality worked but did not provide evidence of how errors were identified and resolved during development.

Documentation and Evaluation

User Documents were generally of a high standard in the majority of samples submitted. The quality of evaluations varied significantly. The higher quality evaluations submitted considered each of the user requirements and assessed how successfully the final system met those requirements. These candidates were able to identify strengths, limitations and possible future developments for the system. Whereas the lower quality evaluations tended to consist of general comments about the system, often lacking supporting evidence. In some cases there was little reference back to the user requirements identified during analysis.

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